



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.

Name of the Course: .Tech (CSE)
Semester: VI
Name of the Paper: Computer Networks – II
Paper Code: TCS614

Time: 1.5 hour

Maximum Marks: 50

Note:

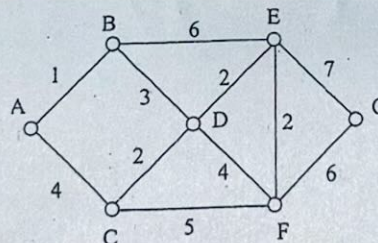
- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)
a. Describe the Route Information Protocol (RIP) routing in detail. Differentiate between RIP and OSPF protocols. [CO1]

OR

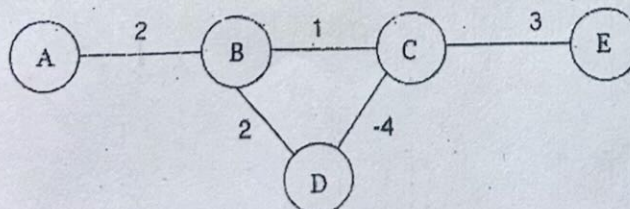
b. Classify the routing protocols. Explain advantages and disadvantages of each class of routing protocols. [CO1]

Q2. (10 Marks)
a. Apply the Link State (LS) Routing algorithm on the given network graph and construct the routing table of router 'A'. [CO1]



OR

b. Apply Distance Vector (DV) routing algorithm on the graph below by selecting 'A' as the starting router. Check whether the DV algorithm is able to find a negative weight cycle present in the graph. [CO1]



Q3. (10 Marks)
a. Explain the detailed working of BGP as an Inter-networking protocol. Also explain the eBGP and iBGP sessions. [CO1]

OR



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- b. Estimate the channel efficiency in pure and slotted ALOHA channels. [CO3]

Q4.

(10 Marks)

- a. Design an algorithm to generate transmitted bits $T(x)$ using Message Polynomial $M(x)$ and Generating Polynomial $G(x)$ by using CRC technique. [CO2]

OR

- b. What are some of the possible services that a data link-layer protocol can offer to the network layer? Which of these data link-layer services have corresponding services in IP? In TCP? [CO2]

Q5.

(10 Marks)

- a. (i) A bit string, 011110111110111110, needs to be transmitted at the data link layer. What is the string actually transmitted after bit stuffing? [CO2]

- (ii) Suppose that a message 1001 1100 1010 0011 is transmitted using Internet Checksum (4-bit word). What is the value of the checksum? [CO2]

OR

- b. What are various services provided by connecting devices (i) Switch and (ii) router. Also differentiate between these two connecting devices.

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